Q2 — Step-by-step Diagrams

This document contains clear, step-by-step diagrams you can copy into your assignment for **Q2(A) Lomuto partition** and **Q2(B) Insertion Sort**.

A) Lomuto partition on `[7, 2, 1, 6, 8, 5, 3, 4]` (pivot = 4)

We'll show the array and the values of `i` and `j` at each compare step. Swaps are shown clearly.

Initial:

```
Index:  0  1  2  3  4  5  6  7
Array: [7, 2, 1, 6, 8, 5, 3, 4]
pivot = A[7] = 4
i = -1
```

Step j = 0 (compare A[0]=7 > pivot)

```
j = 0, A[j] = 7 > 4  --> no swap
i = -1
Array: [7, 2, 1, 6, 8, 5, 3, 4]
```

Step j = 1 (A[1]=2 ≤ pivot)

```
j = 1, A[j] = 2 ≤ 4  --> i <- i+1 = 0; swap A[0] and A[1]
Before swap: [7, 2, 1, 6, 8, 5, 3, 4]
Swap positions 0 and 1
After swap:  [2, 7, 1, 6, 8, 5, 3, 4]
i = 0
```

Step j = 2 (A[2]=1 ≤ pivot)

```
j = 2, A[j] = 1 ≤ 4  --> i <- 1; swap A[1] and A[2]
Before swap: [2, 7, 1, 6, 8, 5, 3, 4]
Swap positions 1 and 2
```

```
After swap:  [2, 1, 7, 6, 8, 5, 3, 4]
i = 1
```

Step j = 3 (A[3]=6 > pivot)

```
j = 3, A[j] = 6 > 4  --> no swap
Array: [2, 1, 7, 6, 8, 5, 3, 4]
i = 1
```

Step j = 4 (A[4]=8 > pivot)

```
j = 4, A[j] = 8 > 4  --> no swap
Array: [2, 1, 7, 6, 8, 5, 3, 4]
i = 1
```

Step j = 5 (A[5]=5 > pivot)

```
j = 5, A[j] = 5 > 4  --> no swap
Array: [2, 1, 7, 6, 8, 5, 3, 4]
i = 1
```

Step j = 6 (A[6]=3 ≤ pivot)

```
j = 6, A[j] = 3 ≤ 4  --> i <- 2; swap A[2] and A[6]
Before swap: [2, 1, 7, 6, 8, 5, 3, 4]
Swap positions 2 and 6
After swap:  [2, 1, 3, 6, 8, 5, 7, 4]
i = 2
```

Final step: place pivot at i+1 = 3

```
Swap A[3] and A[7] (pivot)
Before final swap: [2, 1, 3, 6, 8, 5, 7, 4]
```

```
After final swap:  [2, 1, 3, 4, 8, 5, 7, 6]
Pivot 4 is now at index 3
```

Result (partitioned array):

```
[2, 1, 3, 4, 8, 5, 7, 6]
```

All elements left of index 3 are ≤ 4; all to the right are > 4.

B) Insertion Sort on `A = [8, 3, 10, 1, 6, 4, 2, 9, 7, 5]`

We'll show the array after each pass (after inserting A[i] into sorted prefix A[0..i-1]). Each pass highlights the key element being inserted.

Initial (pass 0):

```
[8, 3, 10, 1, 6, 4, 2, 9, 7, 5]
```

Pass 1 (insert key = 3 at index 1 into [8]):

```
Take key = 3
Compare with 8 → shift 8 right
Array after insertion: [3, 8, 10, 1, 6, 4, 2, 9, 7, 5]
```

Pass 2 (key = 10 at index 2, prefix [3,8]):

```
key = 10
10 ≥ 8 → no shifts
Array: [3, 8, 10, 1, 6, 4, 2, 9, 7, 5]
```

Pass 3 (key = 1 at index 3, prefix [3,8,10]):

```
key = 1
Shift 10 → [3, 8, 10, 10, 6, 4, 2, 9, 7, 5]
Shift 8  → [3, 8, 8, 10, 6, 4, 2, 9, 7, 5]
```

```
Shift 3  → [3, 3, 8, 10, 6, 4, 2, 9, 7, 5]
Place key at index 0 → [1, 3, 8, 10, 6, 4, 2, 9, 7, 5]
```

Pass 4 (key = 6 at index 4, prefix [1,3,8,10]):

```
key = 6
Shift 10 → [1, 3, 8, 10, 10, 4, 2, 9, 7, 5]
Shift 8  → [1, 3, 8, 8, 10, 4, 2, 9, 7, 5]
Place key at index 2 → [1, 3, 6, 8, 10, 4, 2, 9, 7, 5]
```

Pass 5 (key = 4 at index 5, prefix [1,3,6,8,10]):

```
key = 4
Shift 10 → [1,3,6,8,10,10,2,9,7,5]
Shift 8  → [1,3,6,8,8,10,2,9,7,5]
Shift 6  → [1,3,6,6,8,10,2,9,7,5]
Place key at index 2 → [1,3,4,6,8,10,2,9,7,5]
```

Pass 6 (key = 2 at index 6, prefix [1,3,4,6,8,10]):

```
key = 2
Shift 10 → [1,3,4,6,8,10,10,9,7,5]
Shift 8  → [1,3,4,6,8,8,10,9,7,5]
Shift 6  → [1,3,4,6,6,8,10,9,7,5]
Shift 4  → [1,3,4,4,6,8,10,9,7,5]
Shift 3  → [1,3,3,4,6,8,10,9,7,5]
Place key at index 1 → [1,2,3,4,6,8,10,9,7,5]
```

Pass 7 (key = 9 at index 7, prefix [1,2,3,4,6,8,10]):

```
key = 9
Shift 10 → [1,2,3,4,6,8,10,10,7,5]
Place key at index 6 → [1,2,3,4,6,8,9,10,7,5]
```

Pass 8 (key = 7 at index 8, prefix [1,2,3,4,6,8,9,10]):

```
key = 7
Shift 10 → [1,2,3,4,6,8,9,10,10,5]
Shift 9  → [1,2,3,4,6,8,9,9,10,5]
Shift 8  → [1,2,3,4,6,8,8,9,10,5]
Place key at index 5 → [1,2,3,4,6,7,8,9,10,5]
```

Pass 9 (key = 5 at index 9, prefix [1,2,3,4,6,7,8,9,10]):

```
key = 5
Shift 10 → [1,2,3,4,6,7,8,9,10,10]
Shift 9  → [1,2,3,4,6,7,8,9,9,10]
Shift 8  → [1,2,3,4,6,7,8,8,9,10]
Shift 7  → [1,2,3,4,6,7,7,8,9,10]
Shift 6  → [1,2,3,4,6,6,7,8,9,10]
Place key at index 4 → [1,2,3,4,5,6,7,8,9,10]
```

Final sorted array:

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

If you want, I can now: - Export this as a clean PNG or PDF for printing, or - Redraw the diagrams as hand-drawn-style boxes (for a more visual, assignment-ready look).

Tell me which format you prefer and I will generate it next.